Dongwook **Kwon**

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Summary

Highly accomplished AI Research Engineer specializing in Large Language Model (LLM) optimization, Multi-Agent Systems (MAS), and AI-driven anomaly detection for critical infrastructure. Demonstrated research excellence through co-first authored publications in top-tier peer-reviewed venues, including an Oral presentation at EMNLP 2025 and a paper at ACL 2026 (both Industry Tracks), developed in collaboration with LG Electronics, and a first-authored journal article in Data Mining and Knowledge Discovery (Springer Nature) on anomaly detection for industrial control systems in power generation. Inventor of multiple registered patents in network and medical device anomaly detection. A core engineer who engineered and delivered a critical agent-based codebase for a mission-critical vehicle control data project for Hyundai Motor Company. With foundational research experience in nuclear cybersecurity and medical cyber-physical systems, poised to significantly advance the United States' technological edge in critical and emerging fields by engineering scalable, reliable, and secure autonomous AI architectures.

Patents

Registered

Nov 2026 **Categorical Data of Networks Based Network Anomaly Detection Apparatus and Method Thereof**, KR 10-3043922 — *Republic of Korea*

Apr 2026 **Method for Implantable Medical Device to Detect Anomaly in Real Time**, KR 10-2961984 — *Republic of Korea*

Aug 2025 **GPT API-Based Network Anomaly Detection**, KR 10-2848814 — *Republic of Korea*

Pending Applications

Oct 2025 **System for Anomaly Detection Using a Collaborative Multi-AI Determination Module Based on LLM**, KR 10-2025-0160217 — *Republic of Korea*

Publications

Journal Articles

- **D. Kwon**, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, "Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments," *Data Mining and Knowledge Discovery*, vol. 40, art. no. 102, 2026.
- Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, "Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise," *Electronics*, vol. 14, no. 5, p. 890, 2025.

Conference Papers

- R. Chang†, **D. Kwon**†, J. Lee†, and N. Verma, "CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades," in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL), Industry Track (Poster)*, pp. 1329–1340, 2026.
  †Equal contribution.
- J. Lee†, R. Chang†, **D. Kwon**†, H. Singh, and N. Verma, "GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems," in *Proceedings of the EMNLP 2025 Industry Track (Oral)*, pp. 1522–1532, 2025.
  †Equal contribution.
- **D. Kwon**, M. Kim, Y. Kang, J. Lee, and C. Park, "Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers," in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. ***Best Poster Award — Silver.***
- **D. Kwon**, Y. Kang, and C. Park, "Enhancing medical device security with GNN-GRU anomaly detection model," in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. ***Outstanding Paper Award.***
- J. Yang, J. Kim, J. Lee, H. Ryu, **D. Kwon**, S. Yeo, P. Kim, Y. Kim, J. Lim, H. Yoon, and C. Park, "Metaverse: Design of the Car Price Prediction Model Through a Machine-learning Approach," in *Proc. IEEE Int. Conf. Metaverse Computing, Networking and Applications (MetaCom)*, pp. 734–737, 2023.

Research & Professional Experience

Suresoft Technologies — *Seongnam-si, Korea*

AI Engineer, Intelligent Agent Technology Team — *Dec. 2025 - Present*

- Developing an AI validation system utilizing AI agent-based architectures for enterprise model verification.
- Designing autonomous and collaborative AI agents that verify and evaluate machine-learning models at scale.
- Engineered and delivered a critical agent-based codebase for a mission-critical vehicle control data project for Hyundai Motor Company.

Bio-Computing and Machine Learning Laboratory, Kwangwoon University — *Seoul, Korea*

Research Assistant — *Aug. 2021 - Dec. 2025*

- Researched deep-learning algorithms for anomaly detection in complex time-series data, focusing on critical cyber-physical systems including nuclear power generation environments.
- Co-developed an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.

University of Toronto (with LG Electronics Toronto AI Lab) — *Toronto, ON, Canada*

Visiting Graduate Researcher — *Jan. 2025 - Jul. 2025*

- Built an LLM-driven assessment framework for evaluating performance of agentic AI systems.
- Work fed directly into peer-reviewed publications at EMNLP 2025 and ACL 2026.

Qualcomm Institute, University of California San Diego — *San Diego, CA, USA*

AI Development Project Participant, Summer Program — *Jul. 2022 - Aug. 2022*

- Built a personality-based drug-addiction prediction system using machine-learning models.
- Recognized with the program's Achievement Award.

Education

Kwangwoon University — *Seoul, Korea*

Master of Science in Computer Engineering — *Mar. 2024 - Feb. 2026*

- GPA: 4.33/4.5 (98.1%)

University of Toronto — *Toronto, ON, Canada*

Graduate Research Program in Mechanical and Industrial Engineering — *Jan. 2025 - Jul. 2025*

- GPA: 3.68/4.0

Kwangwoon University — *Seoul, Korea*

Bachelor of Science in Electronics and Communications Engineering — *Mar. 2019 - Feb. 2024*

- GPA: 3.41/4.5 (88.1%)

Honors & Awards

Major Awards

Aug. 2022 **Achievement Award**, Qualcomm Institute AI Development Project — UC San Diego — *United States*

Dec. 2021 **Gold Award (Ministerial Award)**, 2021 Smart Maritime Logistics Project — Minister of Oceans and Fisheries — *South Korea*

Academic Recognition

Nov. 2025 **Outstanding Student Paper Award**, Conference of the Institute of Electronics and Information Engineers — *South Korea*

Nov. 2024 **Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC 2024) — *South Korea*

Nov. 2023 **Outstanding Paper Award**, Conference of the Korean Society of Medical and Biological Engineering — *South Korea*